

Zacharie Van Herstraeten

Quantum information & optics researcher

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[Google Scholar](#) • [arXiv](#) • [ORCID](#) • [LinkedIn](#)



Employment

2024 – today	Postdoctoral researcher <i>Inria — École Normale Supérieure (ENS)</i> QAT team, working with Dr. Ulysse Chabaud. Studying the quantum resources underlying bosonic quantum computing via resource-theoretic methods, with a focus on Wigner negativity and entanglement witnesses in continuous-variable systems, alongside foundational questions on the phase-space description of quantum states.	Paris, France
2022 – 2024	Postdoctoral researcher <i>College of Optical Sciences, University of Arizona</i> Quantum Photonic Applications Group of Prof. Saikat Guha. Worked on continuous-variable quantum systems beyond the Gaussian regime, focusing on the role of non-Gaussianity across different settings, with applications to communication capacities, entanglement generation under local operations, and measures of non-classicality.	Tucson, AZ, USA

Education

2017 – 2021	PhD in Quantum Information <i>École polytechnique de Bruxelles, Université libre de Bruxelles (ULB)</i> Center for Quantum Information and Communication (QuIC), supervised by Prof. Nicolas Cerf. Investigated quantum uncertainty in phase space through information theory and majorization, developing entropic tools tailored to continuous-variable quantum systems.	Brussels, Belgium
2015 – 2017	MSc in Physics Engineering <i>École polytechnique de Bruxelles, Université libre de Bruxelles (ULB)</i> Specialized in photonics and quantum applications.	Brussels, Belgium
2015 – 2016	International exchange <i>Norwegian University of Science and Technology (NTNU)</i> Erasmus exchange during the first year of the master's degree.	Trondheim, Norway
2012 – 2015	BSc in Engineering <i>École polytechnique de Bruxelles, Université libre de Bruxelles (ULB)</i> General engineering curriculum.	Brussels, Belgium

Fellowships

2021	BAEF Postdoctoral Fellowship <i>Belgian American Educational Foundation (BAEF)</i> One-year postdoctoral fellowship awarded to support Belgian scholars conducting research in the United States.
2017	F.R.S.-FNRS Doctoral Fellowship (FRIA) <i>Fonds de la Recherche Scientifique (FNRS), Belgium</i> Competitive four-year doctoral fellowship supporting PhD training in the exact and natural sciences.

Academic service

2025 & 2026	Co-organizer <i>QuiDiQua — Quasiprobability Distributions in Quantum Mechanics, Optics and Information</i> Member of the organizing committee of two editions of the international workshop QuiDiQua (Paris, 2025; Brussels, 2026), gathering theorists and experimentalists around quasiprobability distributions in quantum physics.	Paris & Brussels
Since 2025	Package maintainer <i>qopy — Quantum Optics in Python</i> Developer and maintainer of an open-source Python toolkit for Wigner functions, bosonic channels, and phase-space visualization in quantum optics. Available at github.com/zac-vh/qopy .	
Since 2024	Reviewer <i>Peer-reviewed scientific journals</i> Reviewer for <i>Quantum</i> ; <i>Physical Review A</i> ; <i>Quantum Information Processing</i> ; <i>International Journal of Theoretical Physics</i> .	
2017 – 2020	Teaching assistant <i>École polytechnique de Bruxelles, Université libre de Bruxelles (ULB)</i> Taught exercise sessions and lectures in the quantum mechanics course for third-year engineering students. Supervised the master's theses of two students.	Brussels, Belgium